Project Proposal

AMMA

AI-Enabled Maternal Monitoring Assistant

A wearable IoT belt for continuous fetal movement monitoring with AI-driven insights and clinical-grade alert intelligence

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Department	Electronics Engineering — VLSI Design & Technology
Project Domain	Digital Healthcare / Biomedical Engineering / AI-Based Prenatal Monitoring
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Academic Year	2026 – 2027
Estimated Budget	INR 14,000 (Negotiable) (two prototype units + development overhead+ custom design Wearable Belt)

1. Executive Summary

AMMA (AI-Enabled Maternal Monitoring Assistant) is a low-cost, wearable IoT system designed to provide continuous fetal movement monitoring for pregnant women in India. The system consists of a 3D-printed ergonomic abdominal belt housing multiple biosensors, an ESP32-based microcontroller unit for on-device signal processing, a cloud backend for AI-driven trend analysis, and a React-based web dashboard accessible to both mothers and healthcare providers.

The core clinical insight driving this project, validated through direct interaction with an obstetric specialist, is that fetal movement alone is not a diagnostic absolute — rather, deviation from an individual patient's established baseline, tracked over multiple days, is the meaningful signal. AMMA is architected around this insight: it learns each pregnancy's movement baseline, applies circadian-aware thresholds to suppress false alarms, and escalates alerts progressively over a 2–3 hour intervention window — the clinically established response time before a movement decline becomes critical.

2. Problem Statement

Fetal movement counting (kick counting) is one of the most reliable, low-cost indicators of fetal well-being in the third trimester. However, current monitoring methods are manual, subjective,

and episodic — relying entirely on the mother to count and report movements at a scheduled time. This approach has significant gaps:

- No continuous monitoring between antenatal visits, which may be weeks apart
- Manual counting is prone to under-reporting, especially during night-time when the fetus is naturally less active
- A documented clinical case from this project's doctor consultation showed a patient who delayed reporting reduced fetal movement — resulting in discovery of a life-threatening nuchal cord (umbilical cord wrapped around the fetal neck) that could have been averted with earlier detection
- No existing affordable wearable targets the Indian rural or semi-urban pregnant population with a sub-INR 2,000 device cost

AMMA addresses all of these gaps through continuous automated monitoring, intelligent alerting, and cloud-based trend tracking accessible to both the mother and her doctor.

3. Objectives

1. Design and fabricate a comfortable, 3D-printed wearable abdominal belt housing all sensor and electronics components
2. Integrate MPU-6050, piezoelectric transducer, MAX30102, and ESP32 to capture fetal movement, uterine contractions, maternal heart rate, and SpO2
3. Develop ESP32 firmware for real-time signal sampling, threshold-based movement detection, and BLE/Wi-Fi transmission
4. Build a cloud backend (Node.js + Firebase) for data storage, AI-driven trend analysis, and baseline-personalised alert generation
5. Create a web dashboard (React) for mothers and doctors with live monitoring, 7-day trend graphs, and a three-level alert system
6. Validate safety compliance of all sensor emissions against FCC, ICNIRP, AIUM, and WHO guidelines for maternal-fetal use
7. Conduct doctor interaction sessions and iterate on alert logic to minimise false positives and maternal anxiety

4. Hardware Design

4.1 System Architecture Overview

The AMMA system is organised across four layers: the wearable belt (hardware), on-device ESP32 processing, cloud backend, and web dashboard. Data flows from sensors through the ESP32 via I²C and analog GPIO interfaces, then as a JSON payload over Wi-Fi to Firebase, and finally to the React dashboard via Firestore real-time listeners.

4.2 Component List

The following components are used in the AMMA prototype. All components are commercially available in India from suppliers including Robu.in, Robocraze, and Robokits.

No.	Component	Purpose in AMMA	Interface	Safety Note
1	ESP32 DevKit v1 (WROOM-32, 38-pin)	Main MCU — Wi-Fi + BLE, dual-core 240 MHz, ADC/GPIO	USB / UART	Non-ionising RF, <6.3 mW, well below ICNIRP limits
2	MPU-6050 (TDK InvenSense)	6-axis MEMS accel + gyro for fetal movement detection	I²C (400 kHz)	No RF emission; purely mechanical sensing, intrinsically safe
3	Piezoelectric disc transducer (PVDF film, passive)	Detects uterine contractions and low-freq vibration	Analog ADC pin	Passive sensor — emits zero energy, no radiation risk
4	MAX30102 (Maxim Integrated)	Maternal heart rate + SpO2 via PPG optical sensing	I²C	Non-ionising IR/red LED, far below ANSI photobiology limits
5	3.7V 1000 mAh LiPo battery	Primary power source for belt	Direct DC	No RF; follow short-circuit and thermal best practices
6	TP4056 charging module (Type-C, with DW01 protection)	LiPo charge management, over-voltage and under-voltage protection	USB Type-C	DC/chemical only — no RF or ultrasound
7	MT3608 boost converter (3.7V → 5V)	Stable regulated 5V supply for ESP32 from LiPo	DC-DC	EMI negligible at low switching frequency
8	0.96" OLED display (SSD1306, I²C)	Local on-belt status: kick count, HR, battery level	I²C	No significant RF emission
9	Passive buzzer (3–5V)	Local audible alert on device for urgent events	GPIO PWM	Off-body sound — negligible acoustic reaching fetus
10	PCB / perfboard + 2.54mm headers	Prototype mounting and inter-component connections	—	—
11	3D-printed belt enclosure (PLA)	Ergonomic wearable housing — see Section 4.3	—	Non-toxic PLA, skin-safe with silicone liner
12	Elastic waistband + velcro strap	Comfortable adjustable fit across third-trimester sizes	—	—

4.3 3D-Printed Wearable Belt Design

The mechanical housing is a critical component of AMMA. A poorly designed belt will cause discomfort, sensor slippage, and unreliable readings. The belt is designed in three parts: a central electronics module pod, two side-mounted sensor pods, and a flexible elastic waistband that connects all three.

4.3.1 Central Electronics Pod

Dimensions: approximately 90 mm (L) × 55 mm (W) × 22 mm (H). Printed in PLA at 20% infill with 3 perimeter shells for structural rigidity. This pod houses the ESP32 DevKit, TP4056 charger, MT3608 boost converter, OLED display (face-up, flush-fit window in top face), passive buzzer, and the LiPo battery in a separate internally partitioned compartment with a vent slot for thermal dissipation. A recessed USB Type-C cutout on the side allows charging without disassembly. The lid is secured with four M2 screws and a silicone gasket for minimal ingress protection. The OLED window is covered with a 1 mm clear acrylic inlay for scratch protection.

4.3.2 Sensor Pods (Left and Right)

Two smaller pods (approximately 40 mm × 35 mm × 14 mm each) are mounted symmetrically on the waistband, positioned to sit at mid-abdomen on either side of the navel. The left pod houses the MPU-6050 module. The right pod houses the piezoelectric disc (adhered flat against the inner face with medical-grade double-sided tape for skin contact) and the MAX30102 module. All sensor-facing inner surfaces are padded with a 2 mm medical-grade silicone sheet to improve skin contact, reduce noise from ambient vibration, and enhance comfort during extended wear.

4.3.3 Wiring and Strain Relief

I²C bus wires (SDA, SCL, VCC, GND) run through a 6 mm channel moulded into the belt track, protected by braided sleeving. Strain relief loops are printed into the pod entry/exit points to prevent wire fatigue. All wires are approximately 150 mm in length to allow the belt to expand without tension.

4.3.4 Comfort and Fit

The elastic waistband is 60 mm wide and made from breathable spandex-cotton blend fabric, adjustable from 700 mm to 1050 mm circumference to accommodate third-trimester abdominal sizes. The belt is designed for session-based wear of 6–8 hours per day (consistent with the clinician's recommendation that continuous 24-hour wear is unnecessary). All contact surfaces use skin-safe silicone or hypoallergenic fabric. Total weight of the assembled belt is estimated at 120–150 grams — comparable to a standard maternity support belt.

4.3.5 Antenna Placement

Following the RF safety precautionary approach validated in the sensor safety analysis, the ESP32's PCB antenna is oriented away from the abdominal surface — positioned at the top face of the central pod — to maximise distance between the 2.4 GHz BLE transmitter and the fetal tissue. BLE transmission is configured in short-burst mode (100 ms advertisement window every 10 seconds) rather than continuous broadcast.

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5. Firmware Architecture (ESP32)

The ESP32 firmware is developed using the Arduino framework within PlatformIO IDE. The firmware implements a dual-core task architecture using FreeRTOS, which the ESP32 supports natively.

5.1 Core 0 — Sensor Sampling

- MPU-6050 polled at 50 Hz via I²C. A rolling window of 20 samples computes peak acceleration magnitude. Any burst exceeding a configurable threshold (default: 0.4g above baseline) is classified as one fetal kick event and appended to the session counter.

- Piezo ADC sampled at 100 Hz on ADC1_CH0. A software high-pass filter (cutoff: 1 Hz) removes DC offset. Contraction events are flagged when sustained pressure elevation exceeds 3 seconds.
- MAX30102 polled at 10 Hz on the same I²C bus (address 0x57). SparkFun MAX3010x library provides heart rate (BPM) and SpO2 (%) derived from red and IR photoplethysmogram signals.

5.2 Core 1 — Connectivity and Display

- OLED updated every 2 seconds with current kick count, heart rate, battery voltage, and Wi-Fi status icon.
- JSON payload constructed every 30 seconds and POSTed to the Firebase REST API endpoint over Wi-Fi. Payload includes: timestamp, session kick count, per-minute kick rate, heart rate, SpO2, piezo contraction flag, battery level.
- BLE advertisement mode activates automatically when Wi-Fi is unavailable. A companion phone app (future scope) or direct browser BLE API can relay data to the cloud.
- Deep-sleep mode engages if no movement is detected for 20 consecutive minutes and the mother has not manually activated monitoring, reducing current draw to under 50 µA.

6. Cloud Backend Architecture

6.1 Technology Stack

- Firebase Cloud Firestore — document store for patient profiles and timestamped sensor readings
- Firebase Cloud Functions (Node.js 18) — serverless REST API for ESP32 data ingestion and scheduled trend analysis
- Firebase Authentication — secure login for mothers (patient role) and doctors (provider role) with role-based Firestore security rules
- Firebase Hosting — static hosting for the React web dashboard

6.2 Data Model

Each patient document in Firestore contains a sub-collection of readings (one document per 30-second session) and a computed baseline document updated daily. The baseline document stores the patient's rolling 7-day average kick count per session and a standard deviation, enabling the AI engine to express alerts as personalised z-score deviations rather than population-level thresholds.

6.3 AI Trend Engine

A Cloud Function scheduled to run hourly performs the following analysis for each active patient:

8. Fetches all readings from the last 72 hours from Firestore
9. Computes per-session kick frequency (kicks per hour) and movement intensity (peak acceleration per kick)

10. Fits a linear trend to the last 7 sessions' kick frequency to detect a rising or declining slope
11. Compares the latest session value to the patient's personalised baseline using a z-score. z > -1.5 = Normal, -1.5 to -2.5 = Watch, < -2.5 = Urgent
12. Applies circadian correction: sessions between 22:00 and 06:00 use a wider normal band (baseline - 40%) to account for fetal sleep state alignment with maternal circadian rhythm
13. Writes the computed alert level and trend direction back to the patient document for the dashboard to display in real time

7. Web Dashboard (React)

7.1 Mother's Dashboard

- Live monitoring panel: current session kick count, live heart rate graph (Recharts), SpO2 value, and battery level
- Daily summary card showing today's kicks vs. personal baseline with a simple colour-coded indicator
- 7-day trend chart (line graph) with a shaded normal range band derived from the patient's own history
- Alert banner using reassurance-first language: green = 'Your baby has moved [n] times — within your normal range'; yellow = 'Movement is below your usual baseline — observe and rest'; red = 'Significant movement drop detected — please contact your healthcare provider now'
- Session history log with timestamps and downloadable PDF report for antenatal appointments

7.2 Doctor's Dashboard

- Patient list view showing each patient's name, gestational week, last active session timestamp, and current alert level badge
- Per-patient detail view with full 14-day movement trend, contraction event log, and ability to annotate observations
- Configurable alert thresholds per patient — the system supports a high-risk mode (future scope) with tighter z-score bands, more frequent monitoring intervals, and SMS/email escalation

7.3 Alert Design Philosophy

Alert design is treated as a patient safety consideration, consistent with the clinical feedback received: poorly designed alerts cause more harm than benefit through anxiety and alarm fatigue. AMMA implements three alert levels (Informational, Advisory, Urgent) with the following rules: a single low-count session never triggers Advisory; two consecutive sessions below the baseline trigger Advisory; a declining trend over 48 hours plus a session more than 2.5 standard deviations below baseline triggers Urgent. All alerts are accompanied by a contextual explanation to reduce anxiety.

8. Sensor Safety Analysis Summary

A detailed sensor safety analysis was conducted and submitted as a separate investigation report. The key findings are summarised below.

Component	Emission type	Measured / estimated SAR	Safety verdict
MPU-6050	None (passive MEMS)	Zero deliberate RF or acoustic output	**Intrinsically safe**
Piezo disc (passive)	None (passive mode)	Zero emission — responds to motion only	**Intrinsically safe**
MAX30102	Non-ionising optical (IR + red LED)	Far below ANSI photobiology limits	**Safe under guidelines**
ESP32 BLE (nRF-class, 2.4 GHz)	Non-ionising microwave RF	~0.0001 W/kg (10,000x below FCC limit)	**Safe — precautionary BLE burst mode applied**
LiPo battery	DC / chemical only	No RF, no ultrasound, negligible static field	**Intrinsically safe**

Mitigation measures incorporated into the design include: BLE burst-mode transmission (100 ms every 10 seconds rather than continuous), ESP32 antenna oriented away from the abdominal surface, TP4056 over-temperature and over-voltage protection, and silicone thermal isolation between battery and skin contact surfaces.

9. Clinical Validation — Doctor Interaction Summary

A structured Q&A session was conducted with an obstetric specialist. Ten clinical questions directly relevant to AMMA's design were asked and answered. The key findings and their design implications are:

Individual baseline tracking (Q1, Q7)

No universal movement pattern exists; individual variation is the norm. AMMA therefore learns each pregnancy's baseline in the first 72 hours rather than using population-level thresholds. The AI engine prioritises frequency (how often movement occurs) and intensity (peak acceleration per movement) as the primary quantifiable metrics.

2–3 hour intervention window (Q2)

Regardless of whether movement reduction is gradual (placental insufficiency) or sudden (cord compression), a 2–3 hour window typically exists for medical response. AMMA's alert escalation timeline is calibrated to this window: Advisory alerts trigger at the start of the window, Urgent alerts with a direct call-to-action trigger at the midpoint.

Multi-parameter analysis (Q3)

Movement alone is insufficient for complete fetal health prediction. Future scope includes integration of maternal SpO2 trend, contraction pattern (from the piezo channel), and eventually Doppler-based fetal heart rate. Current implementation explicitly notes this limitation in the dashboard UI.

Circadian-aware thresholds (Q4, Q10)

Night-time movement reduction is physiologically normal. AMMA's alert engine applies adjusted thresholds between 22:00 and 06:00 to suppress false alerts during fetal sleep states, preventing alarm fatigue.

8-hour session monitoring (Q6)

Continuous 24-hour monitoring is clinically unnecessary and uncomfortable. AMMA supports session-based wear of 6–8 hours, consistent with the clinician's recommendation and directly supporting working women.

Anxiety mitigation (Q9)

Poorly designed alerts cause more harm than benefit. AMMA uses reassurance-first alert language, graduated three-level escalation, and contextual explanations with every alert. The doctor specifically recommended hospital visits for practical demonstrations of fetal heart rate monitoring and maternal/fetal signal differentiation — these are planned for the next project phase.

10. Budget Estimate

The total estimated budget for the AMMA project is INR 14,000(Negotiable), covering all hardware, fabrication, and development expenses for two prototype units. This figure is negotiable based on institutional support and component availability.

Cloud hosting costs are zero — Provided By Our side for the Presentation and Demonstration to guide

11. Future Scope

- Addition of a Doppler-based fetal heart rate sensor for multi-parameter monitoring, directly addressing the clinical finding that movement alone is insufficient for complete fetal health prediction
- High-risk pregnancy mode with tighter alert thresholds, more frequent monitoring intervals, and direct SMS/WhatsApp integration for real-time caregiver notification
- Native Android/iOS companion app to relay BLE data to the cloud in areas without Wi-Fi, and to deliver push notifications for alerts
- Machine learning model trained on de-identified patient datasets to refine baseline estimation and improve false positive suppression
- CDSCO (Central Drugs Standard Control Organisation) medical device registration pathway investigation for commercial deployment under Class B diagnostic device category

- Integration with national maternal health programmes (PMSMA, JSSK) for deployment in public health centres